

BEE 4750 Homework 4: Linear Programming and Capacity Expansion

Name:

ID:

Due Date

Thursday, 11/06/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to analytically (or graphically) solve a linear program.
- Problem 2 asks you to formulate and solve a resource allocation problem using linear programming.
- Problem 3 asks you to formulate, solve, and analyze a standard generating capacity expansion problem.

Load Environment

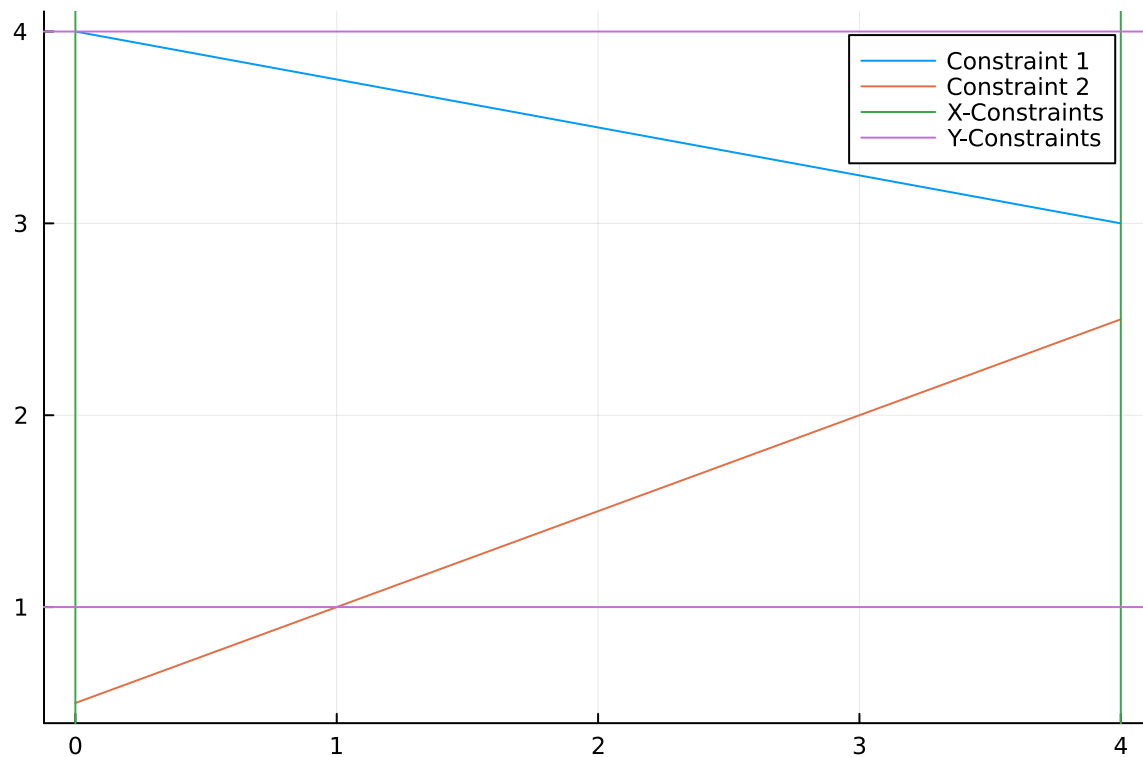
The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
Activating project at `c:\Users\holly\4750 HW\hw4-ellahollyella`
```

```
using JuMP
using HiGHS
using DataFrames
using Plots
using Measures
using CSV
using MarkdownTables
```

```
# Problem 1
m1 = -1/4
b1 = 4
x1 = 0:0.2:4
y1 = m1 .* x1 .+ b1
m2 = 1/2
b2 = 1/2
x2 = 0:0.2:4
y2 = m2 .* x2 .+ b2
plot(x1, y1, label="Constraint 1")
plot!(x2, y2, label="Constraint 2")
vline!([0, 4], label="X-Constraints")
hline!([1, 4], label="Y-Constraints")
```



Based on these results, there are 4 constraint-intersections that are within the realm of possibilities: the points (0,4), (4,3), (1,1), and (4,2.5). The one that yields the max of $3x+2y$ is the point (4,3), making that our solution.

Problems (Total: 30 Points)

Problem 1 (Total: 3 Points)

Solve the following linear program **analytically**. Show your work and justify any reasoning.

$$\begin{aligned}
 &\max_{x,y} && 3x + 2y \\
 &\text{subject to:} && x + 4y \leq 16 \\
 &&& x - 2y \leq -1 \\
 &&& 0 \leq x \leq 4 \\
 &&& 1 \leq y \leq 4.
 \end{aligned}$$

Problem 2 (12 points)

A farmer has access to a pesticide which can be used on corn, soybeans, and wheat fields, and costs \$70/kg-yr to apply. The crop yields the farmer can obtain following crop yields by applying varying rates of pesticides to the field are shown in Table 1.

Application Rate (kg/ha)	Soybean (kg/ha)	Wheat (kg/ha)	Corn (kg/ha)
0	2900	3500	5900
1	3800	4100	6700
2	4400	4200	7900

Table 1: Crop yields from applying varying pesticide rates for Problem 1.

The costs of production, *excluding pesticides*, for each crop, and selling prices, are shown in Table 2.

Crop	Production Cost (\$/ha-yr)	Selling Price (\$/kg)
Soybeans	350	0.36
Wheat	280	0.27
Corn	390	0.22

Table 2: Costs of crop production, excluding pesticides, and selling prices for each crop.

Recently, environmental authorities have declared that farms cannot have an *average* application rate on soybeans, wheat, and corn which exceeds 0.8, 0.7, and 0.6 kg/ha, respectively. The farmer has asked you for advice on how they should plant crops and apply pesticides to maximize profits over 130 total ha while remaining in regulatory compliance if demand for each crop (which is the maximum the market would buy) this year is 250,000 kg?

Problem 2.1

Formulate a linear program for this resource allocation problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations). **Tip: Make sure that all of your constraints are linear.**

Problem 2.2

How many ha should the farmer dedicate to each crop and with what pesticide application rate(s)? How much profit will the farmer expect to make?

Problem 2.3

The farmer has an opportunity to buy an extra 10 ha of land. How much extra profit would this land be worth to the farmer? Discuss why this value makes sense and under what conditions you would recommend the farmer should make the purchase.

Problem 2 (15 points)

For this problem, we will use hourly load (demand) data from 2013 in New York's Zone C (which includes Ithaca). The load data is loaded and plotted below in Figure 1.

Problem 2.1: The decision variables are the pesticide application amount and also the hectares for each crop. In order to combine these, we have 9 variables, 3 for each crop to cover the 3 different application rates: s_0 for the ha of soybeans getting 0 application, s_1 for the ha getting 1 kg/ha, and s_2 for those getting 2 kg/ha. The other 2 crops follow suit, resulting in 9 decision variables defined in a 1x9 matrix.

Constraints: We are constrained by total ha (130 max), the regulatory limits for pesticide application for each crop (0.8, 0.7, and 0.6) and the demand for crops (between 0 and 250,000 kg each). If we have variables, s_0 , s_1 , and s_2 , and similarly for w and c , where $s(0-2)$ is total soybean ha (same for other crops), the equations would look as follows:

$$s_0 + s_1 + s_2 + w_0 + w_1 + w_2 + c_0 + c_1 + c_2 \leq 130;$$

$$[(s_0 * 0) + (s_1 * 1) + (s_2 * 2)] / s(0-2) \leq 0.6,$$

$$[(w_0 * 0) + (w_1 * 1) + (w_2 * 2)] / w(0-2) \leq 0.7,$$

$$[(c_0 * 0) + (c_1 * 1) + (c_2 * 2)] / c(0-2) \leq 0.8;$$

However, these are nonlinear. To make them linear, we move the denominator to the other side, resulting in:

$$[(s_0 * 0) + (s_1 * 1) + (s_2 * 2)] \leq 0.6 * s(0-2),$$

$$[(w_0 * 0) + (w_1 * 1) + (w_2 * 2)] \leq 0.7 * w(0-2),$$

$$[(c_0 * 0) + (c_1 * 1) + (c_2 * 2)] \leq 0.8 * c(0-2);$$

As for production:

$$0 \leq (s_0 * 2900) + (s_1 * 3800) + (s_2 * 4400) \leq 250,000, \text{ this will be given the variable name } (Ps) \text{ for future reference.}$$

$$0 \leq (w_0 * 3500) + (w_1 * 4100) + (w_2 * 4200) \leq 250,000, (Pw);$$

$$0 \leq (c_0 * 5900) + (c_1 * 6700) + (c_2 * 7900) \leq 250,000, (Pc);$$

We also know that all of the decision variables must be non-negative.

The goal is to maximize profits, so we need to use the cost of production, cost of pesticide, and the selling point for each crop to develop our objective function:

$$(s_0 + s_1 + s_2)350 + (w_0 + w_1 + w_2)280 + (c_0 + c_1 + c_2)*390 = \text{cost of prod}$$

$$70(1(s_1+w_1+c_1)+2*(s_2+w_2+c_2)) = \text{cost of pesticide}$$

$$P_s * 0.36 + P_w * 0.27 + P_c * 0.22 = \text{revenue}$$

profit = revenue - cost of prod - cost of pesticide, and we are maximizing this

```
# 2.2: opt. code
farm_model = Model(HiGHS.Optimizer) # initialize model object
@variable(farm_model, h[1:9] >= 0) # non-negativity constraints
# total prod for each crop
Ps = (2900*h[1]+3800*h[2]+4400*h[3])
Pw = (3500*h[4]+4100*h[5]+4200*h[6])
Pc = (5900*h[7]+6700*h[8]+7900*h[9])
# total area for each crop
hs = sum(h[1:3])
hw = sum(h[4:6])
hc = sum(h[7:9])

@objective(farm_model, Max, ((Ps*0.36+Pw*0.27+Pc*0.22)-(hs*350+hw*280+hc*390)-(70*(1*(h[2]+h[3]+2*(h[4]+h[5]+h[6]+2*(h[7]+h[8]+h[9]))))))
@constraint(farm_model, total, (sum(h)<=130))
# pesticide amount regulatory constraint
@constraint(farm_model, soybean_app, (0*h[1]+1*h[2]+2*h[3])<=0.8*(hs))
@constraint(farm_model, wheat_app, (0*h[4]+1*h[5]+2*h[6])<=0.7*(hw))
@constraint(farm_model, corn_app, (0*h[7]+1*h[8]+2*h[9])<=0.6*(hc))
# Production max constraint
@constraint(farm_model, soybean_prod, (2900*h[1]+3800*h[2]+4400*h[3])<=250000)
@constraint(farm_model, wheat_prod, (3500*h[4]+4100*h[5]+4200*h[6])<=250000)
@constraint(farm_model, corn_prod, (5900*h[7]+6700*h[8]+7900*h[9])<=250000)
# Production nonnegative constraint
@constraint(farm_model, soybean, (2900*h[1]+3800*h[2]+4400*h[3])>=0)
@constraint(farm_model, wheat, (3500*h[4]+4100*h[5]+4200*h[6])>=0)
@constraint(farm_model, corn, (5900*h[7]+6700*h[8]+7900*h[9])>=0);
```

$$5900h_7 + 6700h_8 + 7900h_9 \geq 0$$

```
optimize!(farm_model)
```

```

Running HiGHS 1.11.0 (git hash: 364c83a51e): Copyright (c) 2025 HiGHS under MIT licence terms
LP   has 10 rows; 9 cols; 36 nonzeros
Coefficient ranges:
  Matrix [2e-01, 8e+03]
  Cost   [7e+02, 1e+03]
  Bound  [0e+00, 0e+00]
  RHS    [1e+02, 2e+05]
Presolving model
7 rows, 9 cols, 27 nonzeros  0s
7 rows, 9 cols, 27 nonzeros  0s
Presolve : Reductions: rows 7(-3); columns 9(-0); elements 27(-9)
Solving the presolved LP
Using EKK dual simplex solver - serial
  Iteration      Objective      Infeasibilities num(sum)
        0      -1.8170297740e+02 Ph1: 7(28.3601); Du: 9(181.703) 0s
        7      -1.1674116702e+05 Pr: 0(0) 0s
Solving the original LP from the solution after postsolve
Model status      : Optimal
Simplex iterations: 7
Objective value    :  1.1674116702e+05
P-D objective error :  6.2325284535e-17
HiGHS run time     :                0.00

```

```

# Results for 2.2
@show value.(h)
@show objective_value.(farm_model)

```

```

value.(h) = [13.812154696132593, 55.248618784530386, 0.0, 6.743306417339566, 15.734381640458]
objective_value.(farm_model) = 116741.16702082448

```

```

116741.16702082448

```

```

# 2.3
@show shadow_price(total);

```

```

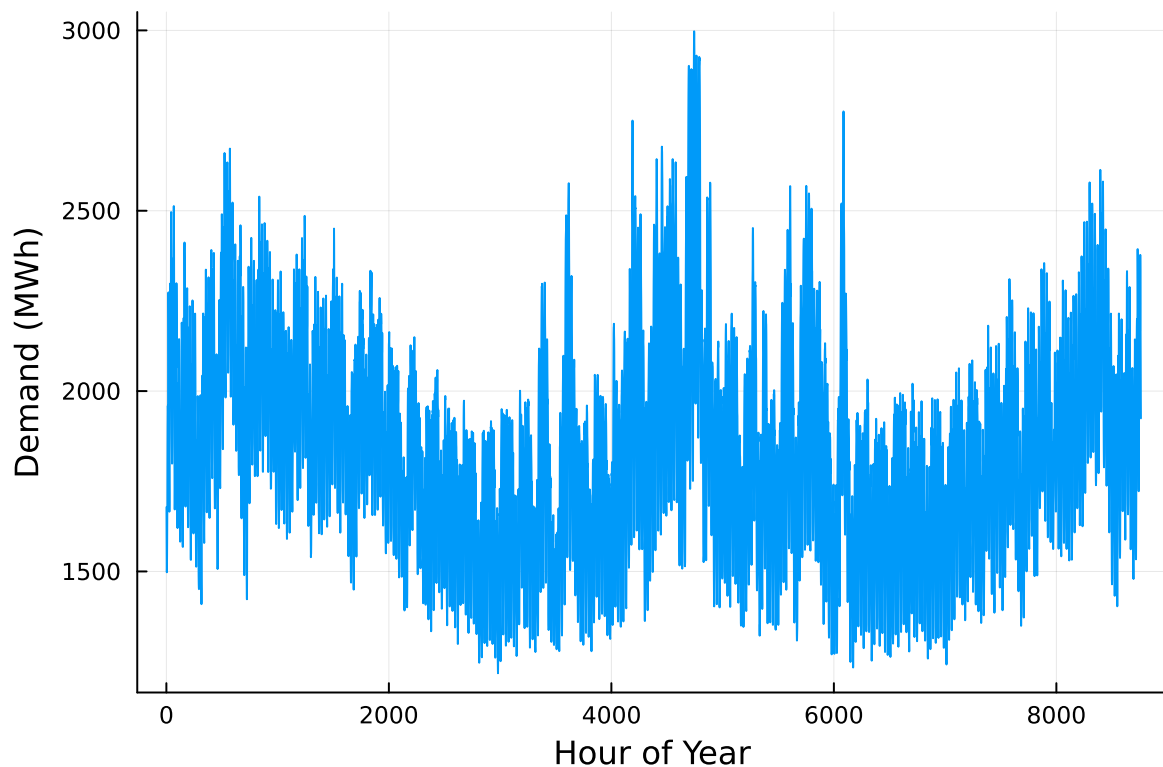
shadow_price(total) = 729.4

```

Since the shadow price for adding 1ha is 729.4, adding 10ha will result in an increase in profit of \$7294.

```
# load the data, pull Zone C, and reformat the DataFrame
NY_demand = DataFrame(CSV.File("data/2013_hourly_load_NY.csv"))
rename!(NY_demand, : "Time Stamp" => :Date)
demand = NY_demand[:, [:Date, :C]]
rename!(demand, :C => :Demand)
demand[:, :Hour] = 1:nrow(demand)

# plot demand
plot(demand.Hour, demand.Demand, xlabel="Hour of Year", ylabel="Demand (MWh)", label=:false)
```



Next, we load the generator data, shown in Table 3. This data includes fixed costs (\$/MW installed), variable costs (\$/MWh generated), and CO₂ emissions intensity (tCO₂/MWh generated).

```
gens = DataFrame(CSV.File("data/generators.csv"))
```


	Plant	FixedCost	VarCost	Emissions
	String15	Int64	Int64	Float64
1	Geothermal	450000	0	0.0
2	Coal	220000	24	1.0
3	NG CCGT	82000	30	0.43
4	NG CT	65000	40	0.55
5	Wind	91000	0	0.0
6	Solar	70000	0	0.0

Finally, we load the hourly solar and wind capacity factors, which are plotted in Figure 2. These tell us the fraction of installed capacity which is expected to be available in a given hour for generation (typically based on the average meteorology).

```
# load capacity factors into a DataFrame
cap_factor = DataFrame(CSV.File("data/wind_solar_capacity_factors.csv"))

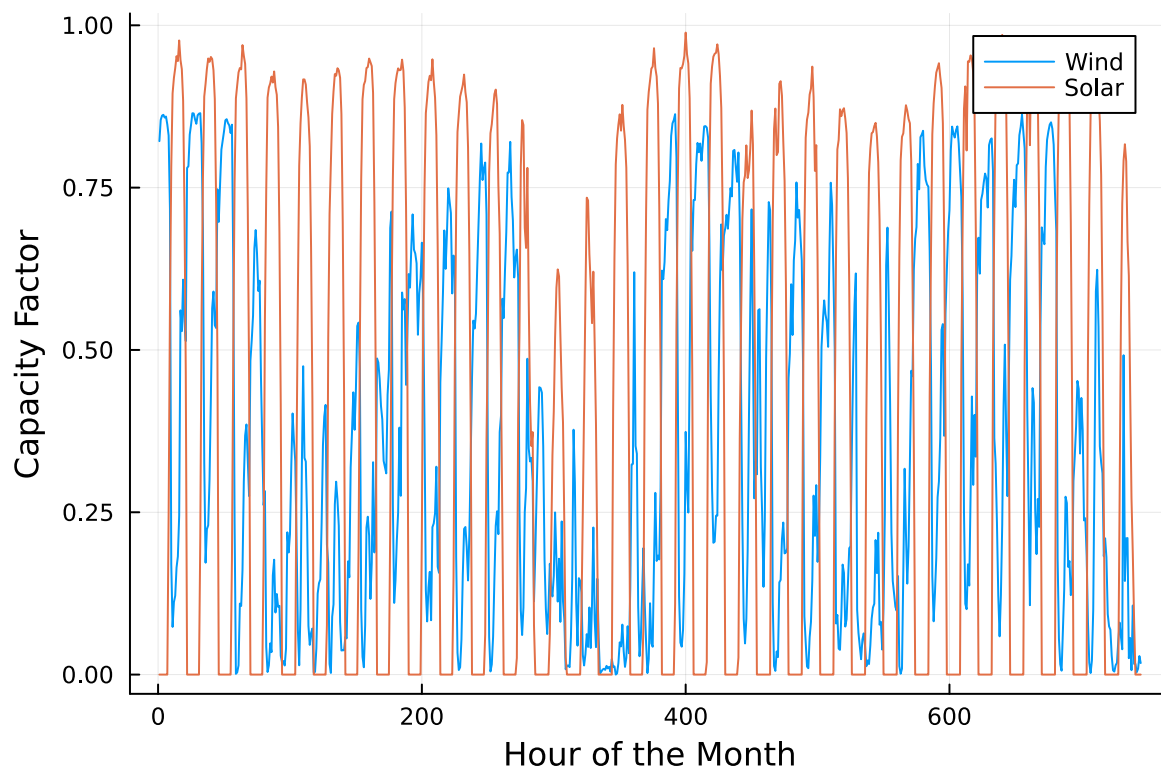
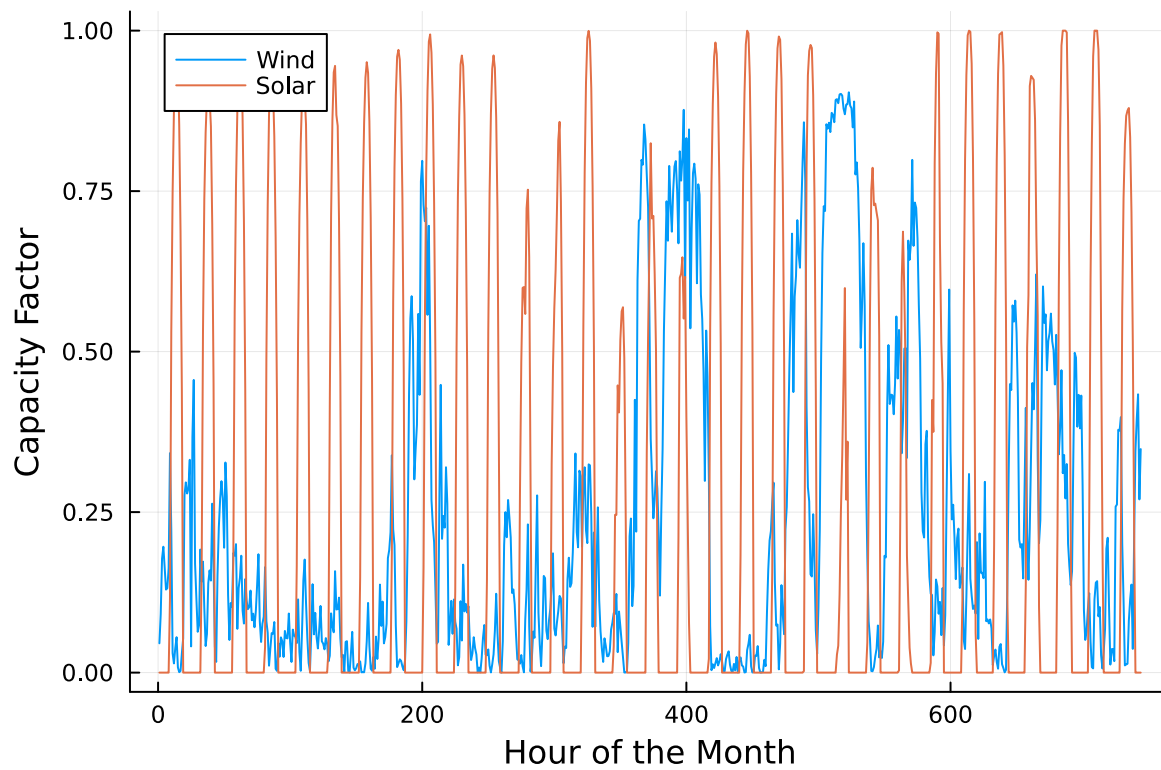
# plot January capacity factors
p1 = plot(cap_factor.Wind[1:(24*31)], label="Wind")
plot!(cap_factor.Solar[1:(24*31)], label="Solar")
xaxis!("Hour of the Month")
yaxis!("Capacity Factor")

# July Cap factors
p2 = plot(cap_factor.Wind[4344:4344+(24*31)], label="Wind")
plot!(cap_factor.Solar[4344:4344+(24*31)], label="Solar")
xaxis!("Hour of the Month")
yaxis!("Capacity Factor")

display(p1)
display(p2)
```

You have been asked to develop a generating capacity expansion plan for the utility in Riley County, NY, which currently has no existing electrical generation infrastructure. The utility can build any of the following plant types: geothermal, coal, natural gas combined cycle gas turbine (CCGT), natural gas combustion turbine (CT), solar, and wind. The NY state legislature is planning to enact an annual CO2 limit, which for the utility would limit the emissions in its footprint to 1.5 MtCO2/yr.

While coal, CCGT, and CT plants can generate at their full installed capacity, geothermal plants operate at maximum 85% capacity, and solar and wind available capacities vary by the hour depend on the expected meteorology. The utility will also penalize any non-served demand at a rate of \$10,000/MWh.



Problem 3.1

Formulate a linear program for this capacity expansion problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations).

Problem 3.2

How much should the utility build of each type of generating plant? What will the total cost be? How much energy will be non-served?

Problem 3.3

What fraction of annual generation does each plant type produce? How does this compare to the breakdown of built capacity that you found in Problem 3.2? Do these results make sense given the demand and generator data?

Problem 3.4

Make a plot of the electricity price in each hour. Discuss any trends that you see.

Problem 3.5

What is the shadow price of CO₂? What is the value to the utility be of allowing it to emit an additional 1000 tCO₂/yr?

Significant Digits

Use `round(x; digits=n)` to report values to the appropriate precision! If your number is on a different order of magnitude and you want to round to a certain number of significant digits, you can use `round(x; sigdigits=n)`.

Getting Variable Output Values

`value.(x)` will report the values of a JuMP variable `x`, but it will return a special container which holds other information about `x` that is useful for JuMP. This means that you can't use this output directly for further calculations. To just extract the values, use `value.(x).data`.

Suppressing Model Command Output

The output of specifying model components (variable or constraints) can be quite large for this problem because of the number of time periods. If you end a cell with an `@variable` or `@constraint` command, I *highly* recommend suppressing

output by adding a semi-colon after the last command, or you might find that your notebook crashes.

References

List any external references consulted, including classmates.

3.1 problem formulation

We assume that the max demand for a given hour is 3000 MWh as can be seen on the provided graph. Our decision variables are the installed capacity of each plant type (MW) (x), the production for each plant (MWh) (y), and the non-served energy (NSE). We want to minimize the total cost, which is the capacity cost based on MW installed, the variable cost (based on production), and the NSE cost, which is \$10,000/MWh.

Constraints: We are constrained by meeting demand, non-negativity, and generation not exceeding capacity for each plant.

```
#3.1 code for LP
cap_factor = DataFrame(CSV.File("data/wind_solar_capacity_factors.csv"))
gens = DataFrame(CSV.File("data/generators.csv"))
NSEcost = 10000
G = 1:nrow(gens)
T = 1:nrow(demand);
# cap factor for renewables, need to include nonrenewables too
# all need to be a vector of length 8760
cf_wind = cap_factor.Wind;
cf_solar = cap_factor.Solar;
cf_geo = fill(0.85, length(cf_wind))
cf_coal = fill(1, length(cf_wind))
cf_ccgt = fill(1, length(cf_wind))
cf_ct = fill(1, length(cf_wind))
# order them how they are presented in a matrix
cf = vcat(
    permutedims(cf_geo),
    permutedims(cf_coal),
    permutedims(cf_ccgt),
    permutedims(cf_ct),
    permutedims(cf_wind),
    permutedims(cf_solar)
)
```

```

renewcap = Model(HiGHS.Optimizer)

# decision vars and non-negativity constraints
@variables(renewcap, begin
    x[g in G] >= 0
    y[g in G, t in T] >= 0
    NSE[t in T] >= 0
end)
@objective(renewcap, Min, sum(gens[G, :FixedCost] .* x) + sum(gens[G, :VarCost] .* sum(y[:, t] for t in T)))
@constraint(renewcap, load[t in T], sum(y[:, t]) + NSE[t] >= demand.Demand[t])
@constraint(renewcap, availability[g in G, t in T], y[g, t] <= x[g]*cf[g, t])

optimize!(renewcap)

```

```

Running HiGHS 1.11.0 (git hash: 364c83a51e): Copyright (c) 2025 HiGHS under MIT licence terms
LP has 61320 rows; 61326 cols; 161976 nonzeros
Coefficient ranges:
  Matrix [5e-05, 1e+00]
  Cost    [2e+01, 4e+05]
  Bound   [0e+00, 0e+00]
  RHS     [1e+03, 3e+03]
Presolving model
56856 rows, 56862 cols, 153048 nonzeros 0s
56853 rows, 56859 cols, 153042 nonzeros 0s
Presolve : Reductions: rows 56853(-4467); columns 56859(-4467); elements 153042(-8934)
Solving the presolved LP
Using EKK dual simplex solver - serial
  Iteration      Objective      Infeasibilities num(sum)
           0      0.0000000000e+00 Pr: 8760(4.1747e+06) 0s
        42008      6.5458487038e+08 Pr: 0(0); Du: 0(2.07029e-10) 5s
Solving the original LP from the solution after postsolve
Model status      : Optimal
Simplex iterations: 42008
Objective value    : 6.5458487038e+08
P-D objective error : 2.4221206776e-14
HiGHS run time     : 5.37

```

```

@show value.(x)
@show objective_value.(renewcap);

```

value.(x) = 1-dimensional DenseAxisArray{Float64,1,...} with index sets:

```
Dimension 1, 1:6
And data, a 6-element Vector{Float64}:
-0.0
 0.0
1658.4514133074133
 879.7538120149645
 484.946167110047
1957.6424239086684
objective_value.(renewcap) = 6.545848703815532e8
```